

Chapter 5: Magnetism and Matter

Formula Sheet + Tips & Tricks

Formula Sheet

Physical Constants Used in This Chapter

Permeability of free space	$\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$
$\frac{\mu_0}{4\pi}$	$= 1 \times 10^{-7} \text{ T m A}^{-1}$
Conversion	$1 \text{ G} = 10^{-4} \text{ T}$

Section I: Magnetic Dipole, Magnetic Moment, etc.

Torque on a Magnetic Dipole

$$\tau = mB \sin \theta$$

m = magnetic moment, B = magnetic field, θ = angle between m and B .

Potential Energy of a Dipole

$$U = -mB \cos \theta = -\vec{m} \cdot \vec{B}$$

Minimum (most stable) when $\theta = 0^\circ$; maximum (unstable) when $\theta = 180^\circ$.

Axial Field of a Short Bar-Magnet

$$B_{\text{axis}} = \frac{\mu_0}{4\pi} \cdot \frac{2m}{r^3}$$

r = distance from centre of magnet along its axis.

Equatorial Field of a Short Bar-Magnet

$$B_{\text{eq}} = \frac{\mu_0}{4\pi} \cdot \frac{m}{r^3}$$

r = distance from centre of magnet along the equatorial line.

Magnetic Moment of an Equivalent Current Loop

$$m = NIA$$

N = number of turns, I = current, A = area of the loop.

Section II: Earth Magnetism

Components of Earth's Magnetic Field

$$B_E = \sqrt{H^2 + V^2}, \quad \tan \delta = \frac{V}{H}$$

H = horizontal component, V = vertical component, δ = angle of dip.

Magnetic Moment of a Uniformly Magnetised Rod

$$m = M \times \text{Volume} = M(\pi r^2 L)$$

M = intensity of magnetisation, r = radius, L = length of the rod.

Permeability and Susceptibility

$$\mu = \frac{B}{H}, \quad \mu_r = \frac{\mu}{\mu_0}, \quad \chi = \mu_r - 1$$

B = magnetic field induction, H = magnetising field, μ_r = relative permeability, χ = magnetic susceptibility.

Intensity of Magnetisation

$$M = \chi H$$

χ = magnetic susceptibility, H = magnetising field.

Curie's Law (Paramagnetic Materials)

$$\chi \propto \frac{1}{T}, \quad \frac{M}{B} \propto \frac{1}{T}$$

T = absolute temperature; susceptibility (and magnetic moment per unit field) rises as T falls.

Tips, Tricks & Hacks

- Bar magnet formulas mirror electric dipole formulas one-to-one: swap $p \leftrightarrow m$ (magnetic moment) and $\frac{1}{4\pi\epsilon_0} \leftrightarrow \frac{\mu_0}{4\pi}$ — if you know Chapter 1's dipole results, you already know most of this chapter.
- Axial field is twice the equatorial field at the same distance, exactly like the electric dipole case.
- Earth's magnetic field has three elements: horizontal component H , dip angle δ

(angle of dip), and declination — $\tan \delta = \frac{V}{H}$ where V is the vertical component; memorising this one relation solves most numericals.

- At the magnetic poles, dip = 90° (field purely vertical); at the magnetic equator, dip = 0° (field purely horizontal).
- Diamagnetic, paramagnetic, ferromagnetic — remember with “Dia repels/pushed out, Para & Ferro attract/pulled in”, and only ferromagnetics retain magnetism (hysteresis) after the field is removed.
- Curie’s law: paramagnetic susceptibility $\chi \propto \frac{1}{T}$ — as temperature rises, paramagnetism weakens (thermal agitation randomises the dipoles).
- Time period of oscillation of a magnet in a magnetic field: $T = 2\pi\sqrt{\frac{I}{mB}}$ — structurally identical to a torsion pendulum/SHM formula, so treat it as SHM.